Identity & Access Domain

DeviceToken

_id: objectId PK
userId: objectId FK
fcmToken: string

deviceType: string ENUM (def: 'android', enum: ios|android|web)

deviceId: string
createdAt: date
updatedAt: date

Permission

_id: objectId PK
name: string UK
resource: string
action: string
description: string
module: string
isActive: boolean (def: true)
isSystem: boolean (def: true)
key: string

status: string ENUM (def: 'active', enum: active|inactive)

createdAt: date
updatedAt: date

Role

_id: objectId PK
name: string UK
slug: string UK
description: string
permissions: objectId[] FK

status: string ENUM (def: 'active', enum: active|inactive)

isSystem: boolean (def: false)
createdBy: objectId FK
updatedBy: objectId FK
createdAt: date
updatedAt: date

Token

_id: objectId PK
user: objectId FK
token: string
expireAt: date
createdAt: date
updatedAt: date

User

_id: objectId PK
name: string
role: string ENUM (def: USER_ROLES.USER, enum:
admin|superAdmin|user|driver)
roleId: objectId FK
email: string UK
profileImage: string (def: '')
dateOfBirth: date

gender: string ENUM (def: GENDER.MALE, enum: male|female)

password: string

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

phone: string
countryCode: string
firebaseUid: string UK
userName: string UK
deviceToken: string
verified: boolean (def: false)

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[] IDX (def: [0, 0])
location_address: string (def: '')
stripeCustomerId: string
referralCode: string UK
referredById: objectId FK

authentication_isResetPassword: boolean (def: false)

authentication_oneTimeCode: number (def: null)
authentication_expireAt: date (def: null)
authentication_authType: string (def: null)
totalCancellations: number (def: 0)
consecutiveCancellations: number (def: 0)
lastCancellationTime: date (def: null)
averageRating: number (def: 0)
totalRatings: number (def: 0)
totalReviews: number (def: 0)
createdAt: date
updatedAt: date

Ride & Booking Domain

CancellationPolicy
_id: objectId PK

passenger_beforeDriverAccepted: passengerScenarioSchema

passenger_afterDriverAccepted: passengerScenarioSchema

passenger_afterDriverArrived: passengerScenarioSchema

driver_afterAccept: driverScenarioSchema
driver_excessiveCancellation: driverScenarioSchema

driver_excessiveCancellationThreshold: number (def: 3)

createdAt: date
updatedAt: date

CancellationReason

_id: objectId PK
reasonName: string
description: string
userType: string ENUM (enum: user|driver)
sortOrder: number (def: 0)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdAt: date
updatedAt: date

driverScenarioSchema

_id: objectId PK
cancellationFee: number (def: 0)
platformShare: number (def: 0)

FareConfiguration
_id: objectId PK
serviceAreaId: objectId FK IDX
serviceCategoryId: objectId FK IDX
rideCategoryId: objectId FK IDX
baseFare: number (def: 0)
perKmFare: number (def: 0)
perMinuteFare: number (def: 0)
waitingFeePerMinute: number (def: 0)
minimumFare: number (def: 0)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdBy: objectId FK
createdAt: date
updatedAt: date

passengerScenarioSchema
_id: objectId PK
cancellationFee: number (def: 0)
platformShare: number (def: 0)
driverCompensation: number (def: 0)

PeakHour

_id: objectId PK
name: string
startTime: string
endTime: string
timezone: string
applicableDays: string[] ENUM (enum:
monday|tuesday|wednesday|thursday|friday|saturday|sunday)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdAt: date
updatedAt: date

Ride

_id: objectId PK
userId: objectId FK IDX
driverId: objectId FK IDX
carId: objectId FK
serviceAreaId: objectId FK
serviceCategoryId: objectId FK
rideCategory_categoryId: objectId FK
rideCategory_name: string
rideCategory_commissionRate: number
pickup_address: string

pickup_location_type: string ENUM (def: 'Point', enum: Point)

pickup_location_coordinates: number[] IDX
destination_address: string
destination_location_type: string ENUM (def: 'Point', enum:
Point)
destination_location_coordinates: number[] IDX
routeInfo_totalDistanceKm: number
routeInfo_totalDurationMinutes: number
routeInfo_googleRouteId: string
routeInfo_polyline: string
driverMatching_requestExpireSeconds: number
driverMatching_searchRadiusKm: number
driverMatching_requiredDriverCount: number
rideType: string ENUM (def: RIDE_TYPE.INSTANT, enum:
instant|scheduled)
scheduledAt: date
timezone: string
reservationStatus: string ENUM (def: 'pending', enum:
pending|confirmed|cancelled|expired)
reservationConfirmedAt: date
reservationAssignedAt: date
reservationAcceptedAt: date
reservationExpiresAt: date
assignedDriverId: objectId FK IDX
reservationReminderSent__24h_: boolean (def: false)
reservationReminderSent__1h_: boolean (def: false)
reservationReminderSent__30m_: boolean (def: false)
reservationReminderSent__15m_: boolean (def: false)
reservationCancelledReason: string
shareToken: string UK
isSharingActive: boolean (def: false)

status: string IDX ENUM (enum:
searching_driver|driver_accepted|waiting_user_approval|driver_on_the_

pickupVerification_method: string ENUM (enum:
otp|phone_last_4_digits)
pickupVerification_otp_code: string
pickupVerification_otp_expiresAt: date
pickupVerification_otp_createdAt: date

pickupVerification_otp_verified: boolean (def: false)

pickupVerification_otp_verifiedAt: date
pickupVerification_otp_attempts: number (def: 0)

pickupVerification_phoneLastFourDigits_value: string

pickupVerification_phoneLastFourDigits_verified: boolean (def:
false)

pickupVerification_phoneLastFourDigits_verifiedAt: date

dropVerification_method: string ENUM (enum:
otp|phone_last_4_digits)
dropVerification_otp_code: string
dropVerification_otp_expiresAt: date
dropVerification_otp_createdAt: date
dropVerification_otp_verified: boolean (def: false)
dropVerification_otp_verifiedAt: date
dropVerification_otp_attempts: number (def: 0)
dropVerification_phoneLastFourDigits_value: string
dropVerification_phoneLastFourDigits_verified: boolean (def:
false)

dropVerification_phoneLastFourDigits_verifiedAt: date

fare_baseFare: number
fare_distanceFare: number
fare_timeFare: number
fare_stopWaitingCharge: number
fare_cancellationFee: number
fare_discount: number
fare_subtotal: number
fare_commission: number
fare_driverEarning: number
fare_total: number
fare_surgeMultiplier: number (def: 1.0)
fare_surgeApplied: number (def: 0.0)
fare_rideFare: number
fare_pendingCancellationFee: number
payment_method: string ENUM (enum:
stripe|wallet|apple_pay|google_pay|card)

payment_status: string ENUM (enum: pending|paid|failed|refunded)

payment_stripePaymentIntentId: string
payment_stripeCheckoutSessionId: string
payment_paidAt: date

cancellation_cancelledBy: string ENUM (enum: user|driver|admin)

cancellation_cancellationReasonId: objectId FK
cancellation_cancellationReasonName: string
cancellation_cancellationFee: number
cancellation_driverCompensation: number
cancellation_platformShare: number
cancellation_cancellationPolicy_scenario: string
cancellation_cancellationPolicy_policyName: string

cancellation_cancellationPolicy_cancellationFee: number

cancellation_cancellationPolicy_driverCompensation: number

cancellation_cancellationPolicy_platformShare: number

cancellation_paymentStatus: string ENUM (enum:
pending|paid|voided)
cancellation_paymentCollectionMode: string
cancellation_rideStatusBeforeCancellation: string

cancellation_surgeSnapshot_multiplier: number (def: 1.0)

cancellation_surgeSnapshot_amount: number (def: 0.0)

cancellation_fareSnapshot: mixed
cancellation_cancelledAt: date
requestedAt: date IDX (def: Date.now)
acceptedAt: date
userApprovedAt: date
arrivedAt: date
startedAt: date
completedAt: date
createdAt: date
updatedAt: date

Ride_NotifiedDrivers
_id: objectId PK
driverId: objectId FK
sentAt: date
respondedAt: date

status: string ENUM (enum: sent|accepted|rejected|expired)

Ride_Stops
_id: objectId PK
order: number
address: string

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[]
isCompleted: boolean (def: false)
completedAt: date

Ride_VerificationAttempts
_id: objectId PK
method: string ENUM (enum: otp|phone_last_4_digits)
attemptedAt: date (def: Date.now)
success: boolean
ipAddress: string

RideCategory
_id: objectId PK
serviceCategoryId: objectId FK IDX
name: string
description: string
commissionRate: number
minimumDriverRating: number
vehicleRequirements: vehicleRequirementSchema

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

supportsReservation: boolean (def: true)
reservationFee: number (def: 0)
createdBy: objectId FK
createdAt: date
updatedAt: date

SurgeRule
_id: objectId PK
ruleName: string UK
ruleType: string IDX ENUM (enum:
default_surge|airport_surge|event_surge|peak_hour_surge|holiday_surg
demandThreshold: number
supplyThreshold: number
minMultiplier: number (def: 1.0)
maxMultiplier: number (def: 1.0)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdBy: objectId FK
createdAt: date
updatedAt: date

Tracking
_id: objectId PK
rideId: objectId FK IDX
driverId: objectId FK
userId: objectId FK
driverLocation_type: string ENUM (enum: Point)
driverLocation_coordinates: number[] IDX
userLocation_type: string ENUM (enum: Point)
userLocation_coordinates: number[] IDX
lastUpdatedAt: date (def: Date.now)
remainingDistanceKm: number
estimatedArrivalMinutes: number
etaCalculatedAt: date
lastDriverLocationUpdateAt: date
driverOnTheWayAt: date
driverArrivedAt: date
targetIsPickup: boolean
targetIsDestination: boolean
currentStopOrder: number (def: -1)

targetType: string ENUM (enum: pickup|stop|destination)

targetLocation_type: string ENUM (def: 'Point', enum: Point)

targetLocation_coordinates: number[] (def: [0, 0])
targetStopOrder: number (def: null)
activeLeg_from: string
activeLeg_to: string
activeLeg_distanceKm: number
activeLeg_durationMinutes: number
activeLeg_isCurrent: boolean
totalDistanceKm: number
totalDurationMinutes: number
polyline: string
createdAt: date
updatedAt: date

Tracking_RouteLegs
_id: objectId PK
from: string
to: string
distanceKm: number
durationMinutes: number
isCurrent: boolean

vehicleRequirementSchema
_id: objectId PK
vehicleTypes: string[]
minimumSeats: number
luggageCapacity: number (def: 0)

Driver & Fleet Domain

airportQueuePrioritySchema
_id: objectId PK
enabled: boolean (def: false)
priorityPosition: number (def: 0)

bonusMultiplierSchema
_id: objectId PK
enabled: boolean (def: false)
multiplierValue: number (def: 1.0)

Car

_id: objectId PK
driverId: objectId FK IDX
brand: string
model: string
year: number
carType: string
seatNumber: number
licensePlate: string
vin: string (def: '')
vehicleLicense: string (def: '')
personalAutoInsurance: string (def: '')
isVerified: boolean (def: false)
verifiedAt: date (def: null)
createdAt: date
updatedAt: date

Car_InsuranceHub
_id: objectId PK
fileUrl: string
fileName: string (def: '')
uploadedAt: date (def: null)
extractedData: mixed (def: {})

DestinationFilter

_id: objectId PK
driverId: objectId FK
tierId: objectId FK
destination_address: string

destination_location_type: string ENUM (def: 'Point', enum: Point)

destination_location_coordinates: number[]
coordinates: number[] IDX
arrivalTime: date
radiusKm: number (def: 5)
status: string ENUM (def: 'ACTIVE', enum:
ACTIVE|EXPIRED|CANCELLED|COMPLETED)
remainingQuota: number
activatedAt: date (def: Date.now)
expiredAt: date (def: null)
cancelledAt: date (def: null)
createdAt: date
updatedAt: date

destinationFilterSchema

_id: objectId PK
enabled: boolean (def: false)
dailyLimit: number (def: 0)

Driver

_id: objectId PK
userId: objectId FK

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[] IDX (def: [0, 0])
location_address: string (def: '')
stripeConnectedAccountId: string
isStripeOnboarded: boolean (def: false)
liveSelfie: string (def: '')
drivingLicense: string (def: '')
ssn: string (def: '')
serviceAreaId: objectId FK (def: null)
serviceAreaAssignedAt: date (def: null)
serviceAreaChangedAt: date (def: null)
taxVerificationStatus: string ENUM (def:
VERIFICATION_STATUS.PENDING, enum:
pending|processing|verified|rejected)
taxVerified: boolean (def: false)
taxVerifiedAt: date (def: null)
taxClassification: string ENUM (def:
CLASSIFICATION.INDIVIDUAL, enum:
individual|sole_proprietorship|llc|partnership|corporation|nonprofit|other)
taxLegalName: string (def: '')
taxBusinessName: string (def: '')
taxIdType: string ENUM (def: TAX_ID_TYPE.SSN, enum:
ssn|itin|tin|ein)
taxIdValue: string (def: '')
taxEmail: string (def: '')
taxPhone: string (def: '')
taxStreet: string (def: '')
taxCity: string (def: '')
taxState: string (def: '')
taxZipCode: string (def: '')
taxCountry: string (def: '')
receiveTaxDocumentsDigitally: boolean (def: true)
driverAvailabilityStatus: string ENUM (def:
DRIVER_A

VAILABILITY_STATUS.OFFLINE, enum:
online|offline|on_trip|break)
lastOnlineAt: date (def: null)
lastOfflineAt: date (def: null)
totalCancellations: number (def: 0)
consecutiveCancellations: number (def: 0)
lastCancellationTime: date (def: null)
averageRating: number (def: 0)
totalRatings: number (def: 0)
totalReviews: number (def: 0)
averageAppreciation: number (def: 0)
totalAppreciationReceived: number (def: 0)
totalAppreciationAmount: number (def: 0)
availability_canReceiveRide: boolean (def: true)

availability_blockedReason: string ENUM (def: null, enum:
daily_limit|continuous_limit|break_required|manual_block)

availability_blockedUntil: date (def: null)
currentPoints: number (def: 0)
lifetimePoints: number (def: 0)
currentTier: objectId FK (def: null)
nextTier: objectId FK (def: null)
progressPercentage: number (def: 0)
tierAchievedAt: date (def: null)
approvalStatus: string ENUM (def: DRIVER_STATUS.PENDING,
enum: none|pending|approved|rejected)
backgroundCheckStatus: string ENUM (def:
VERIFICATION_STATUS.PENDING, enum:
pending|processing|verified|rejected)

identityVerificationStatus: string ENUM (def:
VERIFICATION_STATUS.PENDING, enum:
pending|processing|verified|rejected)
licenseExpiryDate: date (def: null)
suspension_isSuspended: boolean (def: false)
suspension_suspendedBy: objectId FK (def: null)
suspension_suspendedAt: date (def: null)
suspension_reason: string (def: '')
suspension_note: string (def: '')
createdAt: date
updatedAt: date

Driver_CancellationHistory
_id: objectId PK
rideId: objectId FK
cancellationReasonId: objectId FK
cancellationReasonName: string
cancelledAt: date (def: Date.now)
cancellationFee: number
platformShare: number
driverCompensation: number
pointsDeducted: number
cancellationPolicy_scenario: string
cancellationPolicy_policyName: string
cancellationPolicy_cancellationFee: number
cancellationPolicy_driverCompensation: number
cancellationPolicy_platformShare: number

Driver_RecentDestinations
_id: objectId PK
title: string (def: '')
placeId: string (def: '')

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[] IDX (def: [0, 0])
location_address: string (def: '')
lastVisitedAt: date (def: Date.now)
createdAt: date (def: Date.now)

Driver_TaxDocuments
_id: objectId PK
documentType: string ENUM (enum:
ssn_card|itin_letter|ein_letter|w9|w8ben|other)
fileUrl: string
fileName: string (def: '')
extractionStatus: string ENUM (def:
EXTRACTION_STATUS.PENDING, enum:
pending|processing|completed|failed)
extractedData: mixed (def: {})
uploadedAt: date (def: Date.now)

DriverDutyPolicy
_id: objectId PK
name: string

scopeType: string ENUM (enum: global|state|city|zone|airport)

countryId: objectId FK (def: null)
stateId: objectId FK (def: null)
cityId: objectId FK (def: null)
zoneId: objectId FK (def: null)
airportId: objectId FK (def: null)
maxDrivingHoursPerDay: number
maxContinuousDrivingHours: number
breakAfterHours: number
breakDurationMinutes: number
maxTripsPerDay: number
minimumRestHours: number

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdAt: date
updatedAt: date

DriverPointHistory
_id: objectId PK
driverId: objectId FK
rideId: objectId FK (def: null)
referralId: objectId FK (def: null)
transactionId: objectId FK (def: null)

eventType: string ENUM (enum:
ride_completed|five_star_rating|airport_ride|scheduled_ride|peak_hour_

source: string
points: number
balanceAfter: number
notes: string
createdBy: objectId FK (def: null)
ruleId: objectId FK (def: null)
referenceId: objectId (def: null)

action: string ENUM (def: null, enum: earning|deduction)

previousBalance: number (def: null)
balanceChange: number (def: null)
newBalance: number (def: null)
metadata_type_rideId: objectId FK
metadata_type_reportId: objectId FK
metadata_type_referralId: objectId FK
metadata_type_adminId: objectId FK
metadata_type_notes: string
metadata_type_source: string
createdAt: date
updatedAt: date

PointRule

_id: objectId PK
name: string

eventType: string UK ENUM (enum:
ride_completed|five_star_rating|airport_ride|scheduled_ride|peak_hour_

points: number
actionType: string ENUM (enum: earning|deduction)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

isDeleted: boolean (def: false)
createdAt: date
updatedAt: date

premiumRideAccessSchema
_id: objectId PK
enabled: boolean (def: false)
allowedCategories: string[] (def: [])

priorityDispatchSchema
_id: objectId PK
enabled: boolean (def: false)
boostMultiplier: number (def: 1.0)

reservationAccessSchema

_id: objectId PK
enabled: boolean (def: false)
maxAdvanceHours: number (def: 0)

ServiceArea

_id: objectId PK
country: string (def: '')
state: string (def: '')
city: string (def: '')
zone: string (def: '')
airport: string (def: '')
countryId: objectId FK (def: null)
stateId: objectId FK (def: null)
cityId: objectId FK (def: null)

type: string IDX ENUM (enum: state|city|zone|airport|country)

maxDrivers: number (def: 0)

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[] (def: [0, 0])
coverageRadiusKm: number (def: 25)
timezone: string (def: 'UTC')

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdAt: date
updatedAt: date

ServiceCategory
_id: objectId PK
name: string UK
description: string
image: string (def: '')

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

supportsReservation: boolean (def: true)
minimumAdvanceBookingMinutes: number (def: 30)
maximumAdvanceBookingDays: number (def: 30)
createdAt: date
updatedAt: date

Tier

_id: objectId PK
name: string UK
code: string UK
level: number UK
requirements: tierRequirementsSchema
benefits: tierBenefitsSchema (def: () =>
DEFAULT_TIER_BENEFITS)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

isDeleted: boolean (def: false)
createdAt: date
updatedAt: date

tierBenefitsSchema

_id: objectId PK
destinationFilter: destinationFilterSchema (def: () =>
DEFAULT_TIER_BENEFITS.destinationFilter)
priorityDispatch: priorityDispatchSchema (def: () =>
DEFAULT_TIER_BENEFITS.priorityDispatch)
reservationAccess: reservationAccessSchema (def: () =>
DEFAULT_TIER_BENEFITS.reservationAccess)
premiumRideAccess: premiumRideAccessSchema (def: () =>
DEFAULT_TIER_BENEFITS.premiumRideAccess)

airportQueuePriority: airportQueuePrioritySchema (def: () =>
DEFAULT_TIER_BENEFITS.airportQueuePriority)

bonusMultiplier: bonusMultiplierSchema (def: () =>
DEFAULT_TIER_BENEFITS.bonusMultiplier)
vipSupport: vipSupportSchema (def: () =>

DEFAULT_TIER_BENEFITS.vipSupport) TierHistory

_id: objectId PK
driverId: objectId FK
oldTierId: objectId FK (def: null)
newTierId: objectId FK
points: number
reason: string
createdAt: date
updatedAt: date

tierRequirementsSchema
_id: objectId PK
pointsRequired: number (def: 0)
tripsRequired: number (def: 0)
ratingRequired: number (def: 0)
acceptanceRateRequired: number (def: 0)

vipSupportSchema
_id: objectId PK
enabled: boolean (def: false)
supportLevel: string (def: 'basic')

Finance & Payment Domain

Payout
_id: objectId PK
payoutId: string UK IDX
userId: objectId FK IDX
amount: number
currency: string (def: 'USD')
status: string IDX ENUM (def: PAYOUT_STATUS.PENDING, enum:
pending|processing|completed|failed|rejected)
method: string ENUM (enum: stripe|bank_transfer)
destinationAccountId: string
transactionId: objectId FK IDX
failureReason: string
processedAt: date
createdAt: date
updatedAt: date

PendingPayment
_id: objectId PK
userId: objectId FK
rideId: objectId FK
type: string ENUM (def: 'cancellation_fee', enum:
cancellation_fee|driver_appreciation)
driverId: objectId FK
amount: number

status: string ENUM (def: 'pending', enum: pending|paid|voided)

paidWithRideId: objectId FK
driverCompensation: number
platformShare: number
stripeSessionId: string
checkoutSessionExpiresAt: date
paymentAttemptCount: number (def: 0)
createdAt: date
updatedAt: date

Transaction

_id: objectId PK
transactionId: string UK IDX
userId: objectId FK IDX
driverId: objectId FK IDX
bookingId: objectId FK IDX
rideId: objectId FK IDX
walletId: objectId FK IDX
stripeCustomerId: string
stripeCheckoutSessionId: string
stripePaymentIntentId: string
stripeChargeId: string
stripeTransferId: string
stripePayoutId: string
stripeRefundId: string
amount: number
totalFare: number
commission: number
fee: number
currency: string
status: string
paymentMethod: string ENUM (enum:
stripe|wallet|apple_pay|google_pay|card)

paymentStatus: string IDX ENUM (def: PAYMENT_STATUS.PENDING, enum: pending|paid|failed|refunded)

transactionType: string IDX ENUM (enum:
booking_payment|wallet_topup|refund|cancellation_fee|cancellation_com

gatewayTransactionId: string IDX
gatewayResponse: mixed
description: string
metadata: mixed
createdAt: date
updatedAt: date

Wallet

_id: objectId PK
userId: objectId UK FK IDX
balance: number (def: 0)
currency: string (def: 'USD')
status: string ENUM (def: WALLET_STATUS.ACTIVE, enum:
active|restricted|suspended)
createdAt: date
updatedAt: date

Communication & Engagement Domain

Banner

_id: objectId PK
name: string
description: string
image: string

status: string ENUM (def: 'active', enum: active|inactive)

createdAt: date
updatedAt: date

Broadcast

_id: objectId PK

deliveryType: string ENUM (enum: immediate|scheduled)

title: string
message: string

type: string ENUM (enum:
service_alert|weather_alert|surge_opportunity|maintenance|airport_notic

targetAudience: string ENUM (enum:
all_drivers|all_passengers|by_city|by_state|by_tier)
targetFilters_city: string
targetFilters_state: string
targetFilters_tier: string
targetFilters_userIds: objectId[] FK (def: [])
scheduledAt: date
status: string IDX ENUM (def: BROADCAST_STATUS.PENDING,
enum: pending|scheduled|sent)
sentAt: date
createdBy: objectId FK
createdAt: date
updatedAt: date

Chat

_id: objectId PK
participants: objectId[] FK
lastMessage: objectId FK (def: null)
read: boolean (def: false)
readBy: objectId[] FK
deletedBy: objectId[] FK
isDeleted: boolean (def: false)
status: string ENUM (def: CHAT_STATUS.ACTIVE, enum:
active|deleted)
pinnedMessages: objectId[] FK
communicationType: string ENUM (def:
CHAT_COMMUNICATION_TYPE.OTHER, enum:
regular_ride|scheduled_ride|lost_found|reservation|support|other)
referenceId: objectId
createdAt: date
updatedAt: date

Message
_id: objectId PK
chatId: objectId FK
sender: objectId FK
text: string
image: string (def: '')
read: boolean (def: false)
type: string ENUM (def: MESSAGE_TYPE.TEXT, enum:
text|image|doc|both)
isDeleted: boolean (def: false)
isPinned: boolean (def: false)
pinnedBy: objectId FK
pinnedAt: date
createdAt: date
updatedAt: date

Notification

_id: objectId PK
title: string
text: string
receiver: objectId FK
sender: objectId FK
referenceId_type: objectId

referenceModel: string ENUM (enum: Car|Review|User|Ride|Wallet)

read: boolean (def: false)
type: string ENUM (enum:
super_admin|admin|user|driver|message_new)
createdAt: date
updatedAt: date

NotificationPreference

_id: objectId PK
userId: objectId UK FK IDX
email: boolean (def: true)
push: boolean (def: true)
newRideRequests: boolean (def: true)
rideUpdates: boolean (def: true)
chatMessages: boolean (def: true)
paymentUpdates: boolean (def: true)
promotions: boolean (def: true)
emergencyAlerts: boolean (def: true)
createdAt: date
updatedAt: date

Referral

_id: objectId PK
referrerId: objectId FK IDX
refereeId: objectId UK FK IDX
referralCode: string IDX
referrerRole: string ENUM (enum: user|driver)
status: string ENUM (def: REFERRAL_STATUS.PENDING, enum:
pending|active|completed|expired|cancelled)
rewardAmount: number (def: 0)
rewardStatus: string ENUM (def: REWARD_STATUS.PENDING,
enum: pending|paid)
joinedAt: date (def: Date.now)
completedAt: date

referralType: string ENUM (def: 'USER', enum: USER|DRIVER)

referredUserId: objectId FK
referredDriverId: objectId FK
rewardCurrency: string (def: 'USD')
qualificationProgress: number (def: 0)
qualificationTarget: number (def: 0)
rewardPaid: boolean (def: false)
rewardPaidAt: date
rewardTransactionId: objectId FK
qualificationCompletedAt: date
auditLogs: referralAuditLogSchema[] (def: [])
createdAt: date
updatedAt: date

referralAuditLogSchema
_id: objectId PK
action: string
actor: objectId FK
actorRole: string
details: mixed
timestamp: date (def: Date.now)

Review

_id: objectId PK
rideId: objectId FK IDX
reviewerId: objectId FK IDX
reviewerRole: string ENUM (enum: user|driver)
receiverId: objectId FK IDX
receiverRole: string ENUM (enum: user|driver)
rating: number ENUM (enum: 1|2|3|4|5)
reviewText: string (def: '')
selectedTags: string (def: '')
appreciation: number (def: 0)
status: string ENUM (def: REVIEW_STATUS.ACTIVE, enum:
active|hidden|reported)
rideSnapshot_driverName: string
rideSnapshot_vehicleName: string
rideSnapshot_vehicleNumber: string
rideSnapshot_completedAt: date
rideSnapshot_rideCategory: string
rideSnapshot_pickup: string
rideSnapshot_destination: string
rideSnapshot_fare: number
reviewForId: objectId FK
reviewById: objectId FK
createdAt: date
updatedAt: date

Support & Operations Domain

adminNoteSchema

_id: objectId PK
note: string
adminId: objectId FK
createdAt: date (def: Date.now)

AiAuditLog
_id: objectId PK
action: string IDX
performedBy: objectId FK IDX
userType: string IDX ENUM (enum: admin|driver|system)
details: mixed
createdAt: date
updatedAt: date

AiConversation

_id: objectId PK
driverId: objectId FK IDX
title: string (def: 'New Conversation')
isArchived: boolean IDX (def: false)
createdAt: date
updatedAt: date

AiKnowledge
_id: objectId PK
title: string
module: string IDX
category: string IDX
content: string
searchableContent: string
tags: string[] IDX (def: [])
keywords: string[] IDX (def: [])
language: string (def: 'en')
priority: number (def: 0)
version: number (def: 1)
isActive: boolean IDX (def: true)
aiEnabled: boolean IDX (def: true)
visibility: string IDX ENUM (def: 'driver', enum:
driver|internal|admin_only)
status: string IDX ENUM (def: 'draft', enum:
draft|under_review|published|archived)
allowedRoles: string[] (def: ['driver'])
previousVersionId: objectId FK
publishedAt: date
publishedBy: objectId FK
changeLog: string
isLatest: boolean IDX (def: true)
createdBy: objectId FK
updatedBy: objectId FK
createdAt: date
updatedAt: date

AiSupport
_id: objectId PK
driverId: objectId FK IDX
conversationId: objectId FK IDX
knowledgeIds: objectId[] FK (def: [])
question: string
normalizedQuestion: string
answer: string
aiModel: string
promptVersion: string (def: '1.0')
confidenceScore: number (def: 1.0)
responseStatus: string ENUM (def: 'success', enum:
success|no_match|disabled_module|blocked|error)
responseSource: string ENUM (def: 'knowledge_base', enum:
knowledge_base|fallback)
responseTimeMs: number (def: 0)
tokensUsed: number (def: 0)
language: string (def: 'en')

feedback: string ENUM (def: null, enum: helpful|not_helpful|null)

helpful: boolean (def: false)
adminReviewed: boolean (def: false)
createdAt: date
updatedAt: date

auditLogSchema
_id: objectId PK
action: string
actor: objectId FK
actorRole: string
details: mixed
timestamp: date (def: Date.now)

EmergencyContact
_id: objectId PK
userId: objectId FK IDX
name: string
phone: string
relationship: string
isActive: boolean (def: true)
createdAt: date
updatedAt: date

EmergencyHelpline
_id: objectId PK
callNumber: string
textNumber: string
createdAt: date
updatedAt: date

Faq
_id: objectId PK
question: string
answer: string
createdAt: date
updatedAt: date

LostAndFoundItemCategory
_id: objectId PK
name: string UK

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdAt: date
updatedAt: date

LostFound

_id: objectId PK
rideId: objectId FK IDX
passengerId: objectId FK IDX
driverId: objectId FK IDX
reportNumber: string UK IDX
itemName: string
itemCategory: objectId FK
itemDescription: string
lastSeenLocation: string

reportStatus: string IDX ENUM (def: REPORT_STATUS.REPORTED
reported|under_review|found|not_found|waiting_payment|payment_comp

foundStatus: string IDX ENUM (def: FOUND_STATUS.PENDING,
enum: pending|found|not_found)
recoveryMethod: string ENUM (enum:
passenger_pickup|driver_delivery)

pickupLocation_type: string ENUM (def: 'Point', enum: Point)

pickupLocation_coordinates: number[]
pickupLocation_address: string

deliveryLocation_type: string ENUM (def: 'Point', enum: Point)

deliveryLocation_coordinates: number[]
deliveryLocation_address: string
scheduledAt: date
deliveryFee: number (def: 0)

paymentStatus: string IDX ENUM (def: PAYMENT_STATUS.NOT_REQUIRED, enum:
not_required|pending|paid|failed|refunded)
paymentIntentId: string
paymentTransactionId: objectId FK
paymentReference: string
paymentAmount: number
paymentCurrency: string
passengerConfirmed: boolean (def: false)
driverConfirmed: boolean (def: false)
passengerRated: boolean (def: false)
passengerRating: number
passengerReview: string
adminNotes: string
driverNotes: string
createdBy: objectId FK
auditLogs: auditLogSchema[] (def: [])
createdAt: date
updatedAt: date

LostFound_UploadedFiles
_id: objectId PK
fileUrl: string
fileName: string
uploadedAt: date (def: Date.now)

ReportIssueCategory
_id: objectId PK
issueName: string UK
description: string
estimatedResponseTimeInMinutes: number (def: 60)

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdAt: date
updatedAt: date

Support
_id: objectId PK
userId: objectId FK
name: string
email: string
subject: string
message: string
priority: string ENUM (def: SUPPORT_PRIORITY.LOW, enum:
low|medium|high|urgent)
createdAt: date
updatedAt: date

TripReport
_id: objectId PK
ticketId: string UK IDX
rideId: objectId FK IDX
reporterId: objectId FK IDX
issueId: objectId FK IDX
providedSummaryDetails: string (def: '')
estimatedResponseTimeInMinutes: number (def: 60)
status: string IDX ENUM (def: TRIP_REPORT_STATUS.OPEN,
enum: open|investigating|resolved|rejected)
resolutionNotes: string (def: '')
resolvedBy: objectId FK
resolvedAt: date
rideSnapshot_rideCategory: string
rideSnapshot_pickupAddress: string
rideSnapshot_destinationAddress: string
rideSnapshot_driverId: objectId FK
rideSnapshot_driverName: string
rideSnapshot_vehicleName: string
rideSnapshot_vehicleNumber: string
rideSnapshot_completedAt: date
adminNotes: adminNoteSchema[] (def: [])
auditLogs: auditLogSchema[] (def: [])
createdAt: date
updatedAt: date

System & Configuration Domain

aiSupportConfigSchema
_id: objectId PK
enabled: boolean (def: true)

provider: string ENUM (def: 'google', enum: google|openai)

model: string (def: 'gemini-2.5-flash')
temperature: number (def: 0.2)
maxTokens: number (def: 800)
historyLength: number (def: 5)
enableConversationMemory: boolean (def: true)
minimumConfidence: number (def: 0.5)
allowFallbackAnswer: boolean (def: true)
defaultLanguage: string (def: 'en')

enabledModules: string[] (def: ['Ride', 'Wallet', 'Referral', 'Tier',
'Points', 'Destination Filter', 'Lost Found', 'Support', 'FAQ', 'Documents',
])

suggestedQuestions: string[] (def: ['How do I receive payments?',
'How does Lost & Found work?', 'How do referral rewards work?', 'How
do destination filters work?',])

rateLimit_maxQuestionsPerMinute: number (def: 5)
rateLimit_maxQuestionsPerHour: number (def: 30)
rateLimit_dailyLimit: number (def: 100)

prompts_systemPrompt: string (def: 'You are an AI Support
Assistant for the Alygo platform.

You answer driver queries ONL

Y using
approved platform documentation. Keep answers helpful and brief. If
the query is outside Alygo documentation, politely refuse.')

prompts_fallbackPrompt: string (def: 'I couldn't find an approved
answer for that. Please contact support.')

prompts_safetyPrompt: string (def: 'Never output database
structure, SQL queries, code snippets, internal business policies,
private formulas, passenger secrets, APIs, or internal configurations.')

prompts_noMatchPrompt: string (def: 'I couldn't find an approved
answer for that. Please contact support.')

AuditLog
_id: objectId PK
action: string
performedBy: objectId FK
details: mixed (def: {})
ipAddress: string
userAgent: string
createdAt: date
updatedAt: date

Call

_id: objectId PK
rideId: objectId FK
referenceId: objectId IDX

communicationType: string IDX ENUM (enum:
regular_ride|scheduled_ride|lost_found|reservation|support|other)

channelName: string UK
agoraUidCaller: number
agoraUidReceiver: number
callerId: objectId FK IDX
receiverId: objectId FK IDX
callerRole: string
receiverRole: string

status: string IDX ENUM (def: CALL_STATUS.INITIATED, enum:
initiated|ringing|accepted|connected|ended|rejected|missed|cancelled|fai

callType: string ENUM (def: CALL_TYPE.VOICE, enum: voice)

startedAt: date
answeredAt: date
endedAt: date
durationSeconds: number (def: 0)
ringStartedAt: date
endedBy: objectId FK
endReason: string
missed: boolean (def: false)
rejected: boolean (def: false)
cancelled: boolean (def: false)
failed: boolean (def: false)
networkQuality: string
tokenVersion: string
metadata: mixed (def: {})
createdAt: date
updatedAt: date

driverMatchingSchema
_id: objectId PK
initialSearchRadiusKm: number (def: 5)
radiusExpansionDistanceKm: number (def: 3)
driverVisibilityDurationSeconds: number (def: 60)
rideRequestLifetimeSeconds: number (def: 300)
maxSearchRadiusKm: number (def: 50)

driverReferralConfigSchema
_id: objectId PK
enabled: boolean (def: true)
rewardAmount: number (def: 100)
rewardCurrency: string (def: 'USD')
requiredCompletedTrips: number (def: 10)
qualificationDays: number (def: 30)
payoutDelayHours: number (def: 0)
autoRewardEnabled: boolean (def: true)
maximumRewardsPerDriver: number (def: 10)
shareInstructions: string
termsAndConditions: string
generalNotes: string

driverRewardsConfigSchema
_id: objectId PK
enabled: boolean (def: true)
tierPromotion: boolean (def: true)
autoDowngrade: boolean (def: true)
dailyQuotaResetTime: string (def: '00:00')
timezone: string (def: 'Asia/Dhaka')
destinationFilterRadiusDefault: number (def: 5)

Event

_id: objectId PK
eventName: string
description: string
timezone: string
startDateTime: date
endDateTime: date
serviceAreaId: objectId FK

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[]
coverageRadiusKm: number

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdBy: objectId FK
createdAt: date
updatedAt: date

Holiday
_id: objectId PK
holidayName: string
timezone: string
startDate: date
endDate: date
description: string

status: string ENUM (def: STATUS.ACTIVE, enum: active|inactive)

createdBy: objectId FK
createdAt: date
updatedAt: date

lostFoundConfigSchema
_id: objectId PK
enabled: boolean (def: true)
reportWindowDays: number (def: 7)
maxFiles: number (def: 5)
maxFileSizeMb: number (def: 10)
defaultDeliveryFee: number (def: 0)
returnConfirmationHours: number (def: 48)
autoCloseDays: number (def: 30)

passengerReferralConfigSchema
_id: objectId PK
enabled: boolean (def: true)
rewardAmount: number (def: 20)
rewardCurrency: string (def: 'USD')
qualificationType: string (def: 'rides')
requiredCompletedTrips: number (def: 1)
qualificationDays: number (def: 30)
allowMultipleRewards: boolean (def: false)
maximumRewardsPerUser: number (def: 5)
autoRewardEnabled: boolean (def: true)
shareInstructions: string
rewardTerms: string
generalNotes: string

PlatformSettings
_id: objectId PK
platformName: string (def: 'Alygo')
currency: string ENUM (def: PLATFORM_CURRENCY.USD, enum:
usd)
isMaintenanceMode: boolean (def: false)
supportEmail: string (def: '')
contactNumber: string (def: '')
createdAt: date
updatedAt: date

RecentDestination

_id: objectId PK
userId: objectId FK IDX
address: string
placeName: string

location_type: string ENUM (def: 'Point', enum: Point)

location_coordinates: number[]
lastUsedAt: date IDX (def: Date.now)
useCount: number (def: 1)
createdAt: date
updatedAt: date

referralConfigSchema
_id: objectId PK
passenger: passengerReferralConfigSchema
driver: driverReferralConfigSchema

reservationConfigSchema
_id: objectId PK
enabled: boolean (def: true)
minAdvanceMinutes: number (def: 30)
maxAdvanceDays: number (def: 30)
driverVisibleBeforeMinutes: number (def: 60)
driverAssignmentTimeoutMinutes: number (def: 5)
reminder24h: boolean (def: true)
reminder1h: boolean (def: true)
reminder30m: boolean (def: true)
reminder15m: boolean (def: true)

Rule

_id: objectId PK
content: string
type: string ENUM (enum: privacy|terms|about)
createdAt: date
updatedAt: date

SystemConfiguration
_id: objectId PK
driverMatching: driverMatchingSchema
tracking: trackingSchema
reservation: reservationConfigSchema
lostFound: lostFoundConfigSchema
referral: referralConfigSchema
driverRewards: driverRewardsConfigSchema
aiSupport: aiSupportConfigSchema
createdAt: date
updatedAt: date

trackingSchema
_id: objectId PK
minLocationUpdateIntervalSeconds: number (def: 2)
minMovementDistanceMeters: number (def: 10)
maxGpsAccuracyToleranceMeters: number (def: 50)
arrivalRadiusMeters: number (def: 30)
etaRefreshIntervalSeconds: number (def: 10)
averageSpeedKmh: number (def: 40)
enableSocketOptimization: boolean (def: true)

doc ($lookup)

countryId ($lookup) airportId ($lookup) stateId ($lookup) zoneId ($lookup) cityId ($lookup)

customPolicy ($lookup) globalPolicy ($lookup)

stateDoc ($lookup)

statePolicy ($lookup) cityPolicy ($lookup)

cityDoc ($lookup)

zonePolicy ($lookup)

zoneDoc ($lookup)

airportPolicy ($lookup)

reports ($lookup)

categoryDetails ($lookup)

ride ($lookup)

serviceArea ($lookup) parentCity ($lookup)

user ($lookup)

cars ($lookup)

activeRides ($lookup)

ride ($lookup)

serviceArea ($lookup)

issueCategory ($lookup)

passenger ($lookup) driver ($lookup)

tracking ($lookup)

previousVersionId

publishedBy

createdBy
updatedBy

performedBy

driverId

driverId

conversationId

knowledgeIds

performedBy

targetFilters.userIds

createdBy

rideId

callerId
receiverId

endedBy

passenger passenger passenger.beforeDriverAccepted.afterDriverAccepted.afterDriverArrived

driverdriver.excessiveCancellation.afterAccept

driverId

insuranceHub

participants

lastMessage

readBy
deletedBy

pinnedMessages

userId

serviceAreaId

taxDocuments

recentDestinations

cancellationHistory

rideId

cancellationReasonId

currentTier
nextTier

suspension.suspendedBy

countryId
stateId
cityId
zoneId
airportId

userId

serviceAreaId

createdBy

serviceAreaId

serviceCategoryId

rideCategoryId

createdBy

userId

createdBy

rideId

passengerId
driverId
itemCategory

uploadedFiles

paymentTransactionId

createdBy

auditLogs

actor

chatId

sender

pinnedBy

receiver
sender

userId

userId

transactionId

userId

rideId

driverId

paidWithRideId

userId

referrerId
refereeId

referredUserId

referredDriverId

rewardTransactionId

auditLogs

actor

user

rideId

reviewerId
receiverId

reviewForId
reviewById

userId
driverId

carId

serviceAreaId

serviceCategoryId

rideCategory.categoryId

stops

notifiedDrivers

driverId

assignedDriverId

verificationAttempts

cancellation.cancellationReasonId

serviceCategoryId

vehicleRequirements

createdBy

permissions

createdBy
updatedBy

countryId
stateId
cityId

userId

createdBy

driverMatching

tracking

reservation

lostFound

referral

driverRewards

aiSupport

passenger

driver

driverId

tierId

driverId

rideId

referralId

transactionId

createdBy

ruleId

metadata.type.rideId

metadata.type.reportId

metadata.type.referralId

metadata.type.adminId

requirements

benefits

destinationFilter

priorityDispatch

reservationAccess

premiumRideAccess

airportQueuePriority

bonusMultiplier

vipSupport

driverId

oldTierId
newTierId

rideId

driverId
userId

routeLegs

userId

driverId

bookingId
rideId

walletId

rideId

reporterId

issueId

resolvedBy

rideSnapshot.driverId

adminNotes

auditLogs

adminId

roleId

referredById

userId